

Before starting, make sure you **read all the questions** and understand what is asked of you. Write your answers to the questions in a separate sheet of paper and make sure you **justify yes/no answers**.

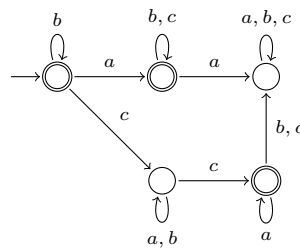
1. (20 points) Translate the following C snippet into LLVM intermediate representation and into MIPS.

```
int ackermann(int x, int y) {
    if (x < 0 || y < 0)
        return -1;
    if (x == 0)
        return y + 1;
    if (y == 0)
        return ackermann(x - 1, 1);
    return ackermann(x - 1, ackermann(x, y - 1));
}
```

2. (6 points) According to the “longest match” principle, what are the first prefixes of

- (a) *aabc*
- (b) *cbcaab*
- (c) *bbbbaa*
- (d) *bcabcaa*
- (e) *bbbcaab*

for which a scanner simulating the following automaton would produce a token?



3. (20 points) Let $T = \{\text{id}, +\}$ be a set of terminals. Is the following grammar LL(0)? Does it define an LL(2) language?

(1)	S	$\rightarrow$	E
(2)	E	$\rightarrow$	$E + E$
(3)		$\rightarrow$	id

4. (12 points) Let $T = \{0, 1, \dots, 9, a, b, \dots, f\}$ be a set of terminals and R be the initial non-terminal symbol. In the following **attribute grammar** the assignments on the right of the grammar rules should be understood as *semantic rules* a.k.a. *attribute functions*.

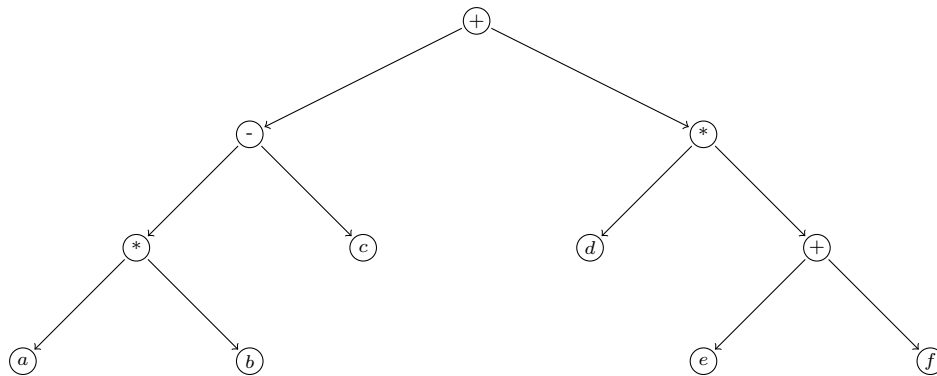
(1)	R	$\rightarrow$	S	$\{d(S) := -1; v(R) := v(S)\}$
(2)	S_0	$\rightarrow$	$S_1 H$	$\{d(S_1), d(H) := d(S_0) + 1; v(S_0) := v(H) + v(S_1)\}$
(3)		$\rightarrow$	H	$\{d(H) := d(S_0) + 1; v(S_0) := v(H)\}$
(4)	H	$\rightarrow$	D	$\{d(D) := d(H); v(H) := v(D)\}$
(5)		$\rightarrow$	A	$\{d(A) := d(H); v(H) := v(A)\}$
(6)	D	$\rightarrow$	0	$\{v(D) := 0\}$
(7)		$\rightarrow$	1	$\{v(D) := 16^{d(D)}\}$
			$\vdots$	
(8)		$\rightarrow$	9	$\{v(D) := 9 \times 16^{d(D)}\}$
(9)	A	$\rightarrow$	a	$\{v(A) := 10 \times 16^{d(A)}\}$
(10)		$\rightarrow$	b	$\{v(A) := 11 \times 16^{d(A)}\}$
			$\vdots$	
(11)		$\rightarrow$	f	$\{v(A) := 15 \times 16^{d(A)}\}$

- (a) (6 points) Classify the attributes, per non-terminal variable, into synthesized and inherited.
 - (b) (6 points) What is the *meaning* (i.e. the semantics) of a word derived from R ?
5. (12 points) During the course we have focused on conservative approximations of both type checking and reachability analysis. Argue that the exact versions of both tasks are in fact undecidable.
6. (20 points) Let $T = \{\text{cst}, \text{id}, +, *\}$ be a set of terminals. Is the following grammar SLR(1)? Is it LALR(2)?

(1)	Exp	$\rightarrow$	Exp + Prod
(2)		$\rightarrow$	Prod
(3)	Prod	$\rightarrow$	Prod * atom
(4)		$\rightarrow$	atom
(5)	atom	$\rightarrow$	cst
(6)		$\rightarrow$	id

Hint: There are no Accept/Reduce or Accept/Shift conflicts, that is to say Accept can co-exist with at most one other action in the action-table cells.

7. (10 points) What is the minimal number of registers required to evaluate the expression tree shown below without using algebraic properties, nor additional memory?



8. (5 points) **Bonus!** Please draw the most helpful depiction of the compilation process that you have seen during this course.